

TDV 2215 Terminal Restart Problem Debug Guide

Problem Analysis

Your terminal is successfully completing the self-test, displaying characters, but then restarting. This indicates an issue in the **main operational loop** rather than the boot sequence.

Root Cause Categories

1. Interrupt System Problems

Most Likely Cause: VSYNC interrupt issues causing system instability

2. Main Loop Corruption

Function Pointer Corruption: Invalid jumps in the main processing loop

3. Timeout/Watchdog Issues

System Timeout: 30-second timeout counter triggering reset

4. Memory Corruption

Stack/Variable Corruption: Corrupted function pointers or stack overflow

Critical Debug Breakpoints

Phase 1: Main Loop Entry Verification

Breakpoint 1.1: `0x2002` (terminal_main_loop entry) **Test:**

- System should reach this point after successful self-test
- Check `system_status_word` (0x5E0C) = 0x8080 (operational state)
- Verify interrupts are enabled

Breakpoint 1.2: `0x20CC` (main loop jump table) **Test:**

- Check `_BYTE_ram_5806` value = 0x20CD
- This is the main character processing dispatch
- **CRITICAL:** If this address is corrupted, system will crash/restart

Phase 2: Character Processing Loop

Breakpoint 2.1: `0x20CD` (character processing function) **Test:**

- This function processes each character input

- Should be called repeatedly during normal operation
- Check parameter values (character codes)

Breakpoint 2.2: End of `(0x20CD)` function (before indirect jump) **Test:**

- Check `(BYTE_ram_5806)` still equals 0x20CD
- If corrupted, this causes restart loop
- Monitor function pointer integrity

Phase 3: Interrupt Coordination

Breakpoint 3.1: `(0x052C)` (vsync_interrupt_handler entry) **Test:**

- Should be called every 16.67ms (60Hz)
- Check `(interrupt_activity_flag)` (0x5E21) behavior
- Monitor `(DAT_ram_5e22)` timeout counter

Breakpoint 3.2: `(0x0634)` (end of VSYNC handler) **Test:**

- `(interrupt_activity_flag)` should be cleared to 0
- Check `(timer_channel_flags)` (0x5E25) processing
- Verify I/O port 0x60 operations

Phase 4: Timeout Monitoring

Breakpoint 4.1: Inside VSYNC handler timeout check **Address:** Around `(0x0580)` (timeout decrement)

Test:

```
c
if ((BYTE_ram_5f0d == 0) && (--DAT_ram_5e22 == 0)) {
    // TIMEOUT CONDITION - May cause restart
    bVar1 = bVar1 | 8; // Set timeout bit
}
```

- Monitor `(DAT_ram_5e22)` countdown (starts at 30000)
- If reaches 0: **System timeout may trigger restart**
- Check `(BYTE_ram_5f0d)` state

Memory Watches During Debug

Critical Function Pointers (Check for Corruption)

Address	Name	Expected Value	Purpose
0x5806	<code>_BYTE_ram_5806</code>	0x20CD	Main loop dispatch
0x5808	<code>_BYTE_ram_5808</code>	0x209D	Secondary dispatch
0x580A	<code>_BYTE_ram_580a</code>	0x20C9	Tertiary dispatch

State Variables (Monitor for Abnormal Values)

Address	Name	Normal Range	Critical Values
0x5E21	<code>interrupt_activity_flag</code>	0x00 or 0x5E	If stuck at 0x5E: Interrupt problem
0x5E22	<code>DAT_ram_5e22</code>	30000 → 0	If 0: Timeout triggered
0x5E25	<code>timer_channel_flags</code>	0x00-0xFF	Active timer channels
0x5DF9	<code>display_refresh_control</code>	0x00-0x0F	Display timing state

System Status

Address	Name	Expected Value	Critical Check
0x5E0C	<code>system_status_word</code>	0x8080	If changed: System state corrupted
0x5F13	<code>MUST_BE_SET_addr_5f13</code>	0x01	Must be 1 for normal operation

Specific Restart Scenarios & Solutions

Scenario 1: VSYNC Interrupt Problem

Symptoms: Restart occurs at regular intervals (16-33ms) **Debug Steps:**

- 1. Set breakpoint at `0x052C` (VSYNC entry)
- 2. Check if interrupt fires every 16.67ms
- 3. Monitor I/O port 0x60 read operations
- 4. Check `expected_control_port2_value_maybe` changes

Common Issues:

- I/O port 0x60 not implemented correctly
- VSYNC interrupt frequency wrong (should be 60Hz)
- Control port status bits incorrect

Fix: Implement proper I/O port 0x60 with expected bit patterns

Scenario 2: Function Pointer Corruption

Symptoms: Restart occurs after several characters processed **Debug Steps:**

1. Watch `_BYTE_ram_5806` for changes from 0x20CD
2. Set memory breakpoint on 0x5806 writes
3. Trace function call stack

Common Issues:

- Stack overflow corrupting memory
- Bad pointer arithmetic in character processing
- Buffer overruns

Fix: Check stack size and memory protection

Scenario 3: 30-Second Timeout

Symptoms: Restart occurs after exactly 30 seconds of operation **Debug Steps:**

1. Monitor `DAT_ram_5e22` countdown in VSYNC handler
2. Check `_BYTE_ram_5f0d` state (should be non-zero to prevent timeout)
3. Set breakpoint when `DAT_ram_5e22` reaches 0

Common Issues:

- Host communication not working (no activity resets timeout)
- `_BYTE_ram_5f0d` incorrectly set to 0
- Timeout handling triggers restart

Fix: Ensure host communication or disable timeout

Scenario 4: Interrupt Flag Stuck

Symptoms: System becomes unresponsive then restarts **Debug Steps:**

1. Monitor `interrupt_activity_flag` (0x5E21)
2. Check if flag gets stuck at 0x5E (not cleared)
3. Verify VSYNC handler clears flag properly

Common Issues:

- Interrupt handler not completing properly
- Main loop not processing interrupt flag
- Race condition between interrupts and main loop

Fix: Ensure proper interrupt/main loop coordination

Quick Diagnostic Test Sequence

Test 1: Basic Operation Check (30 seconds)

1. **Start terminal emulator**
2. **Set breakpoint at 0x2002** (main loop entry)
3. **Continue execution** - should hit immediately
4. **Set breakpoint at 0x20CD** (character processing)
5. **Continue** - should hit when characters are processed
6. **Monitor for 30 seconds** - check for timeout restart

Test 2: Interrupt Coordination Test

1. **Set breakpoint at 0x052C** (VSYNC)
2. **Count hits over 1 second** - should be ~60 hits
3. **Watch** `interrupt_activity_flag` - should toggle 0x00 ↔ 0x5E
4. **Check** `DAT_ram_5e22` - should decrement from 30000

Test 3: Function Pointer Stability Test

1. **Set memory watch on 0x5806** (main dispatch pointer)
 2. **Process several characters**
 3. **Verify pointer remains 0x20CD**
 4. **Check for memory corruption**
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Common Emulator Implementation Issues

1. Missing or Incorrect I/O Port 0x60

Problem: VSYNC handler crashes when reading control port **Fix:** Implement I/O port 0x60 with proper bit patterns

```
c
// Port 0x60 should return control/status bits
// Bits 0-1: Control state
// Other bits: Hardware status
```

2. VSYNC Interrupt Timing Wrong

Problem: Too fast/slow VSYNC causes instability **Fix:** Ensure exactly 60Hz (16.67ms intervals)

3. Stack Overflow

Problem: Insufficient stack space causes corruption **Fix:** Increase stack size or check for infinite recursion

4. Memory Protection

Problem: Function pointers corrupted by bad memory writes **Fix:** Implement memory protection or bounds checking

5. Host Communication Timeout

Problem: No activity causes 30-second timeout restart **Fix:**

- Implement proper host serial communication
 - Or disable timeout by setting `BYTE_ram_5f0d = 1`
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Emergency Fixes for Testing

Disable 30-Second Timeout

Memory Patch: Set `BYTE_ram_5f0d` (0x5F0D) = 1 **Effect:** Prevents timeout-based restart

Force Activity Flag

Memory Patch: Periodically set `interrupt_activity_flag` (0x5E21) = 0x5E **Effect:** Simulates activity to prevent timeout

Bypass Function Pointer Dispatch

Code Patch: Replace indirect jump with direct return **Effect:** Prevents dispatch table corruption issues

Success Indicators

Terminal Running Properly

- **Characters display correctly** ✓ (You have this)
- **No restart after 30+ seconds** ← (Your target)
- **Cursor blinks properly** (1Hz rate)
- **Keyboard input works**
- **Host communication functional**

Use this systematic approach to identify and fix the restart condition in your TDV 2215 terminal emulator.